Department of Electronic Systems

Examination paper for TFE4146 Semiconductor Devices

Examination date: 14.12.2022

Examination time (from-to): 15:00 – 19.00

Permitted examination support material:
C
Specified printed matter:
- Karl Rottmann, Formelsammlung/Collection of mathematical formulas.
- Specified simple calculator

Academic contact during examination: Erik Folven
Phone: 94437094

Academic contact present at the exam location: No

OTHER INFORMATION

Get an overview of the question set before you start answering the questions.

Read the questions carefully and make your own assumptions. If a question is unclear/vague, make your own assumptions and specify them in your answer. Only contact academic contact in case of errors or insufficiencies in the question set. Address an invigilator if you wish to contact the academic contact. Write down the question in advance.

InsperaScan: For question **18 and 19** you are meant to answer on handwritten sheets. Other questions must be answered directly in Inspera. At the bottom of the question you will find a seven-digit code. Fill in this code in the top left corner of the sheets you wish to submit. We recommend that you do this during the exam. If you require access to the codes after the examination time ends, click “Show submission”.

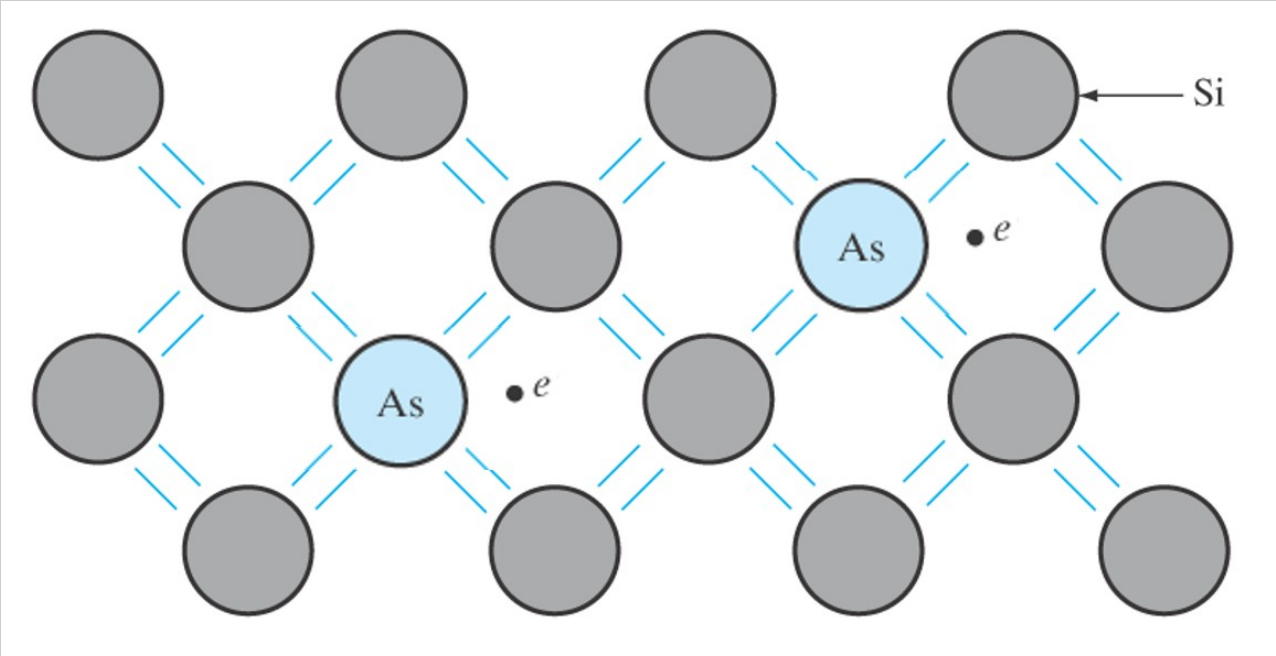
Weighting: The weighting per question is indicated for each question. Blank and wrong answers are typically treated equally in the scoring of multiple choice questions with only one answer, i.e., in these questions there is no risk in guessing an answer if you do not know the correct answer. In questions with more than one alternative there will typically be negative points for wrong/blank answers, but you may not get less than 0p for the question as a whole.

Notifications: If there is a need to send a message to the candidates during the exam (e.g. if there is an error in the question set), this will be done by sending a notification in Inspera. A dialogue box will appear. You can re-read the notification by clicking the bell icon in the top right-hand corner of the screen.

Withdrawing from the exam: If you become ill or wish to submit a blank test/withdraw from the exam for another reason, go to the menu in the top right-hand corner and click “Submit blank”. This cannot be undone, even if the test is still open.

Access to your answers: After the exam, you can find your answers in the archive in Inspera. Be aware that it may take a working day until any hand-written material is available in the archive.

1



The simple schematic above illustrates a Si crystal at room temperature where some of the Si atoms have been replaced by an As atom. In this case, each of the dopant atoms introduces an addtional electron to the crystal.

How will the introduction of As atoms change the net charge of the crystal?
(correct: 2p, wrong: 0p, blank: 0p)
Select one alternative

- The As doping will make the crystal positively charged.
- The As doping will not affect the net charge of the crystal
- The As doping will make the crystal negatively charged.

What do we typically call the kind of doping shown above?
(correct: 2p, wrong: 0p, blank: 0p)
Select one alternative

- Donor doping
- Blood doping
- Acceptor doping
- Intrinsic doping

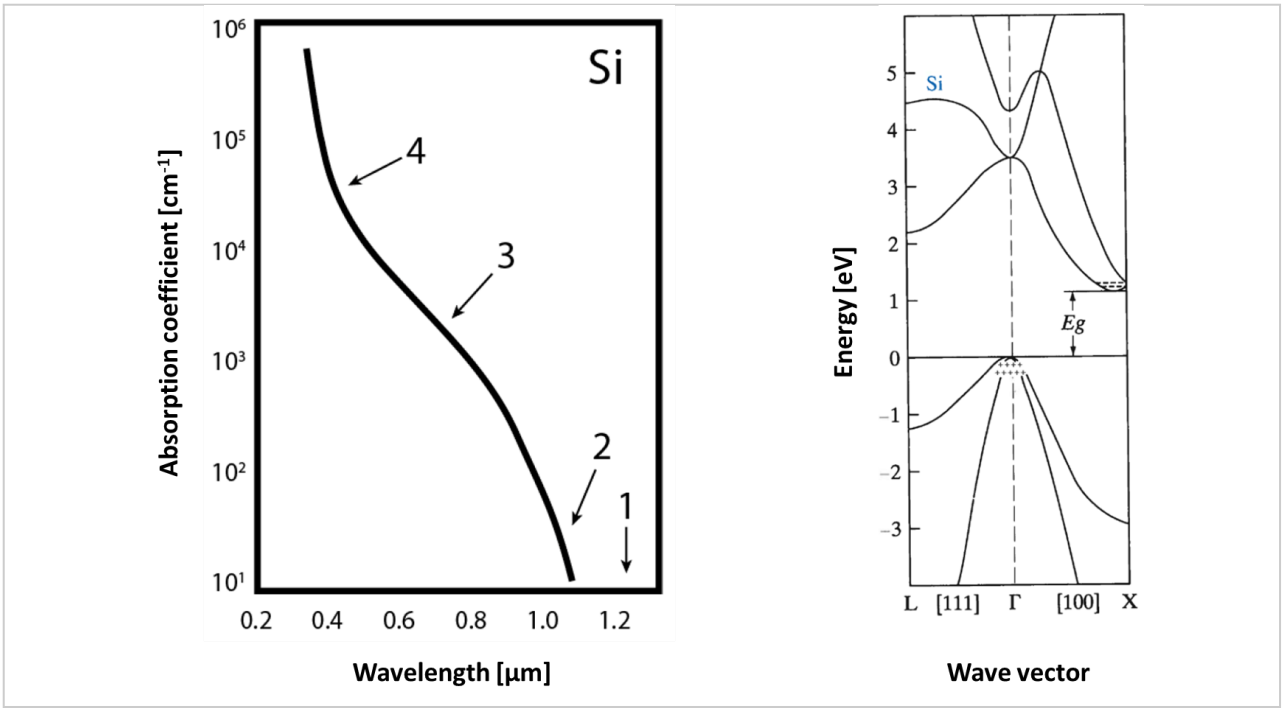
How whould the introduction of As into a Si crystal be represented in a band diagram representation of the crystal?

(correct: 2p, wrong: 0p, blank: 0p)
Select one alternative

- As doping will increase the Fermi energy in the system.
- As doping will reduce the band gap of the system.
- As doping will increase the mobility of the electrons in the system.
- The As doping will increase the valence band energy.

Maximum marks: 6

2



In the above figure it is shown how the optical absorption coefficient, α, for Si depends on the wavelength of incomming light. That is, to how large an extent will light be absorbed by Si at different wavelengths. A small part of the Si bandstructure is also shown. The text below describes the absorption coefficient plot as the wavelength of the incomming light is gradually reduced (going from region 1 to 4).

Fill in the blanks in the following two sections of text. Words are suggested for each blank – choose the most appropriate one. (correct: 1p, wrong: -0.5p, blank: -0.5p, minimum score: 0p)

In region 1 we see that there is virtually no absorption of incoming light. This is because

Select alternative (there is not enough room for a photon with these wavelengths in the first Brillouin zone, Si has six equivalent conduction band minima, the light has lower energy than the indirect band gap in Si, all electron states in the condusction band are filled). As we move to lower

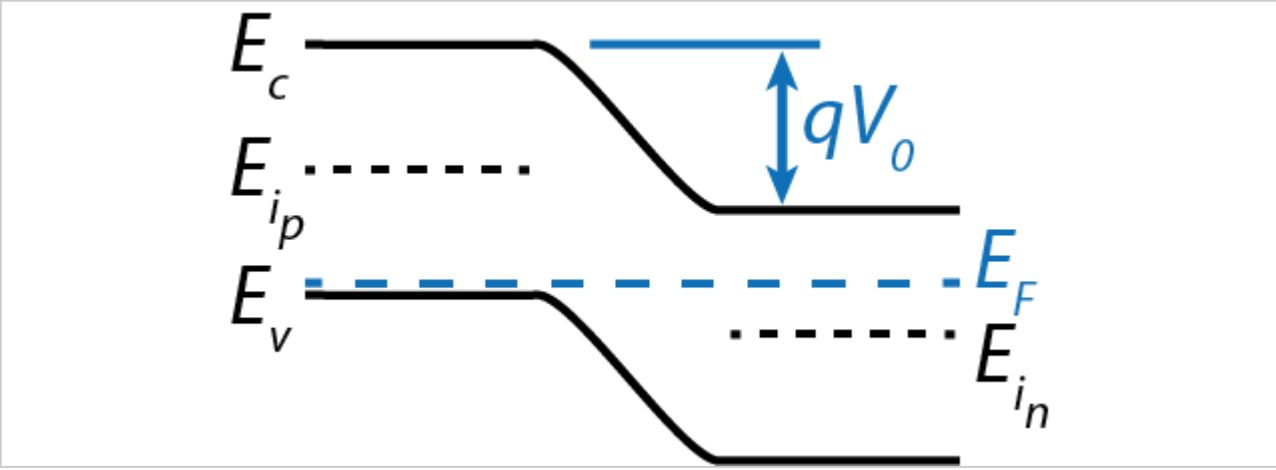
wavelengths in region 2, we see a steep increase in α as Select alternative (the Fermi energy is reduced, the mobility of the electrons is increased, more and more states in the valence band get filled, the energy of the incomming light is now larger than 1.1 eV). The increase in α falls off

somewhat in region 3 as Select alternative (impurity scattering becomes dominant, there is plenty of room at the bottom, the crystall gets hotter and hotter, the growth in the density of states in the conduction band also falls off for higher energies). Finally a second increase in α is seen in

region 4 when the incomming photons Select alternative (have higher energy than the direct band gap of Si, are reflected back and forth between the Brillouin zone boundaries, lights up the entire k-space, reaches further into the valence band due to their high energy).

Maximum marks: 4

3



In the following we will consider a Si p-n junction at room temperature. The band diagram at thermal equilibrium for this junction is shown above.

Fill in the blanks in the following two sections of text. Words are suggested for each blank – choose the most appropriate one. (correct: 1p, wrong: -0.5p, blank: -0.5p, minimum score: 0p)

Within the depletion region in the p-type material, there is a net Select alternative (positive, neutral, negative)charge. This region is depleted of free Select alternative (carriers, acceptors, donors, impurities)and the remaining charge originates from the ionized Select alternative (donors, acceptors, holes, electrons). Moving further away from the junction into the neutral p-type material, we find a large number of free Select alternative (holes, electrons, dopants) and Select alternative (ionized, neutral, positive) acceptors. The concentration of charge carriers is Select alternative (less than, essentially equal to, larger than)that of the acceptors.

In the n-type material, there are electrons and Select alternative (free, neutral, ionized, very few) donors. In the neutral region, the Select alternative (free, immobile, entangled) electrons and the Select alternative (positive and free, neutral and immobile, positive and immobile, neutral and free) donors cancel out

with respect to charge. In the
depleted part of the n-type semiconductor, close to the junction, the semiconductor

has a net Select alternative (negative, positive, neutral) charge. Here, the semiconductor is

depleted of Select alternative (donors, carriers, acceptors) and the

net charge is due to Select alternative (ionized donors, unionized donors, holes, ionized

acceptors). In this particular p-n junction, the doping is highest Select alternative (on the n-side, on the p-side, in the middle, at the gym). As a result, the current at the junction will be

dominated by Select alternative (donors, acceptors, electrons, holes) when the device is under forward bias.

Maximum marks: 14

4

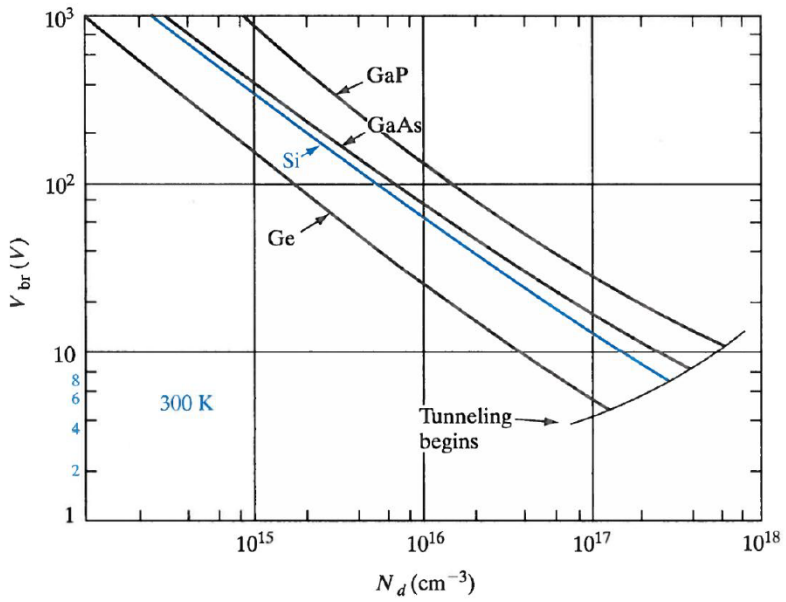


Figure Variation of avalanche breakdown voltage in abrupt p^+-n junctions, as a function of the n-side dopant concentration.

A Si p^+-n junction, 10^{-2} cm^2 in area, has a dopant concentration on the n-side of $N_d=10^{16}cm^{-3}$. The minority carrier lifetime on the n-side is 1µs and the diffusion constant is D_p= 10 cm^2/s. The device is at room temperature (300K) and the intrinsic carrier concentration is $1.5{\cdot}10^{10}$ cm^{-3}. The band gap of Si is 1.1 eV. The relative permittivity for Si is $\epsilon_r = 11.8$.

a) Calculate the current in units µA for a forward bias of 0.5V.
(correct: 5p, wrong: 0p, blank: 0p)

Enter your numerical answer here: [ ] µA.

b) Approximate the value of the contact potential, qV_0, in units eV for this device.
(correct: 5p, wrong: 0p, blank: 0p)

Enter your numerical answer here: [ ] eV.

Hint: It may be helpful to sketch the band diagram for this device to make the correct assumptions in your calculations.

c) Given the figure above, what is the depletion region width W in units µm at the onset of avalanche breakdown in this junction?
(an approximate value from the figure above is sufficient to arrive at an acceptable answer)
(correct: 5p, wrong: 0p, blank: 0p)

Enter your numerical answer here: [ ] µm

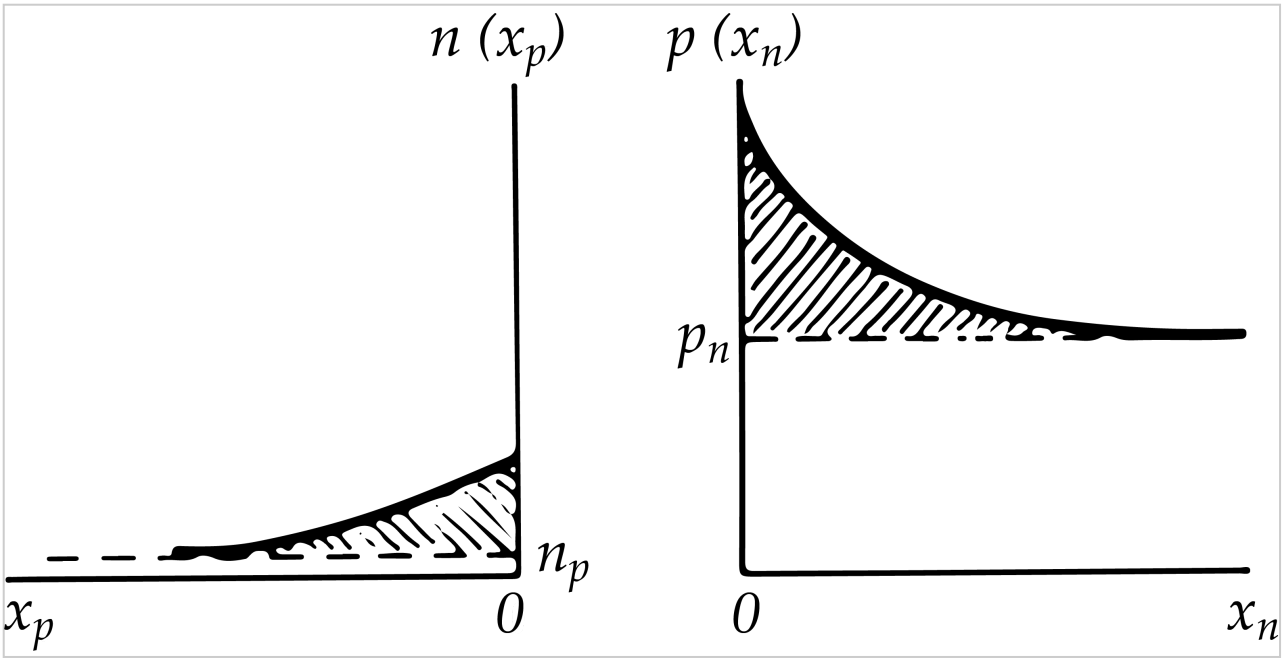
Maximum marks: 15

5 Consider a one-sided p+-n junction with a forward bias voltage $V_b >> \frac{kT}{q}$. What happens to the different current components if the bias voltage is increased by a factor 2?
(correct: 2p, wrong: -2p, blank: 0p, minimum score 0p)
Select one or more alternatives:

- [ ] The drift of holes increase by a factor 2 and the drift of electrons decrease by a factor 2.
- [ ] The drift of holes and electrons remain approximately unchanged
- [ ] The diffusion of holes and electrons increase by a factor 2
- [ ] The diffusion of holes increase exponentially, the electron current remain unchanged
- [ ] The diffusion of holes and electrons increase exponentially
- [ ] The diffusion of holes increase exponentially and the diffusion of electrons decrease exponentially.
- [ ] The drift of holes and electrons increase by a factor 2

Maximum marks: 4

6

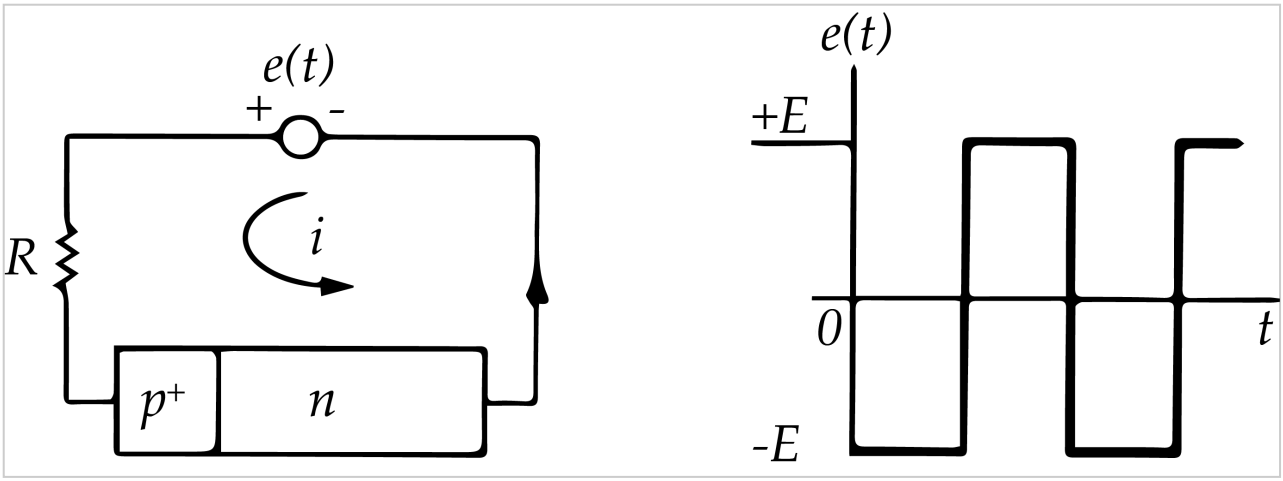


The figure above shows the minority carrier distribution in which of the following devices?
(correct: 2p, wrong: 0p, blank: 0p)
Select one alternative:

- A p-n junction under forward bias in which the p side is the most heavily doped.
- A p-n junction under forward bias in which the n side is the most heavily doped.
- A p-n junction under reverse bias in which the n side is the most heavily doped.
- A p-n junction under reverse bias in which the p side is the most heavily doped.

Maximum marks: 2

7



Consider the circuit sketched above containing a voltage source, a resistor and a p^+-n junction. The voltage delivered from the voltage source as a function of time is showed in the plot to the right. If the current just before switching the voltage at *t=0* is equal to *I* , what is true for the current measured at a time $t=t_1$ when you assume that t_1 is small compared to both the period of the input signal from the voltage source and the minority carrier lifetime on the n-side of the p^+-n junction.

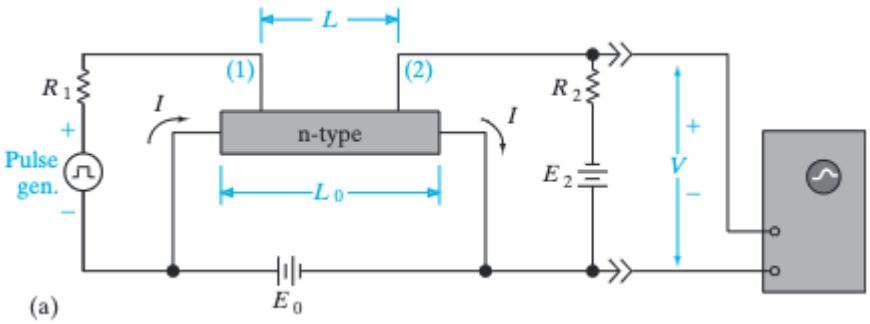
(correct: 2p, wrong: 0p, blank: 0p)

Select one alternative:

- The current at t_1 is approximately equal to *I*
- The current at t_1 is approximately equal to 0
- The current at t_1 is approximately equal to 2*I*
- The current at t_1 is approximately equal to -*I*

Maximum marks: 2

8



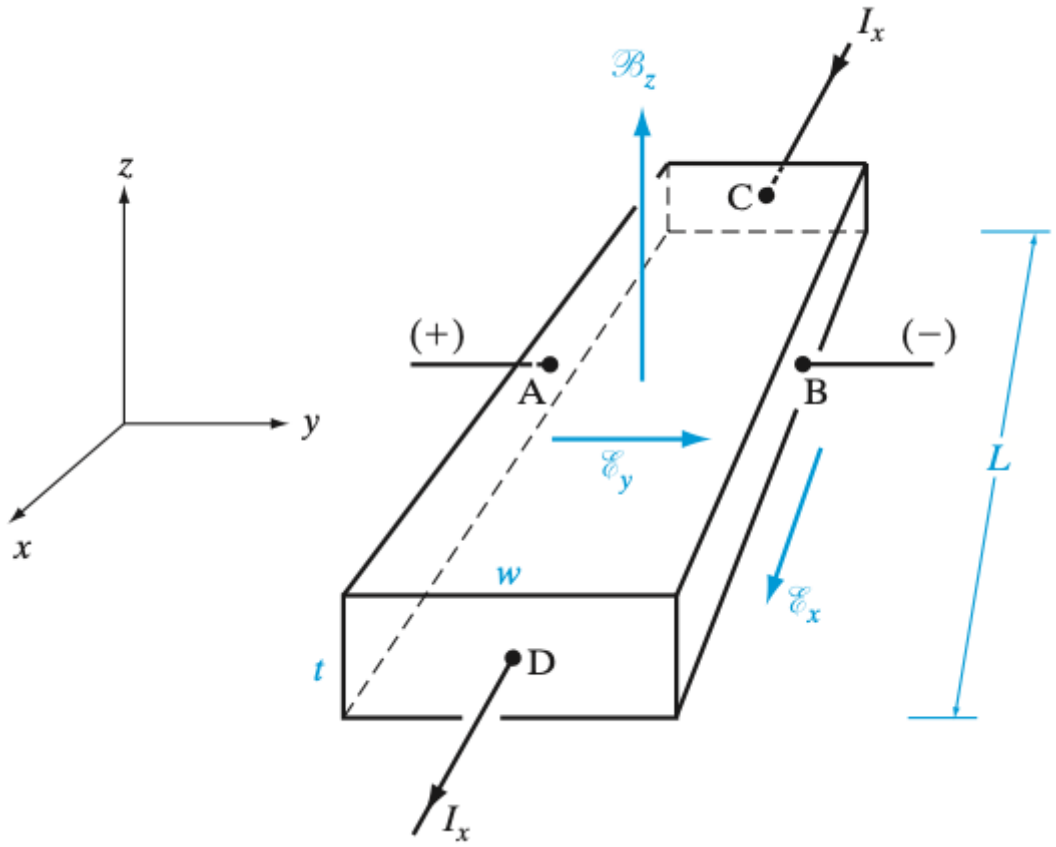
The figure above shows the experimental setup for the Haynes-Shockley experiment.
Which parameters can be measured using the Haynes-Shockley experiment?
(correct: 2p, wrong: -2p, blank: 0p, minimum score 0p)

Select one or more alternatives:

- [ ] majority carrier concentration
- [ ] diffusion coefficient for majority carriers
- [ ] mobility of majority carriers
- [ ] mobility of minority carriers
- [ ] minority carrier concentration
- [ ] diffusion coefficient for minority carriers

Maximum marks: 4

9



The figure above shows the setup for the Hall experiment. The n-type semiconductor bar has a width $w = 0.01\text{cm}$, thickness $t = 10^{-3}\text{cm}$ and length $L = 0.5\text{cm}$. A magnetic field of $B_z = 10^{-4}\text{Wb/cm}^2$ is applied in the direction shown. For a current $I_x = 1\text{mA}$ we have $V_{AB} = -2\text{mV}$ and $V_{CD} = 100\text{mV}$.

Given: The force on a particle with charge Q and velocity $\vec{v}$, from a magnetic field $\vec{B}$, is $\vec{F} = Q\vec{v} \times \vec{B}$

Find the concentration and mobility of the majority carrier and fill in the numerical answer below.

Majority carrier concentration : $\cdot 10^{17}\text{cm}^{-3}$

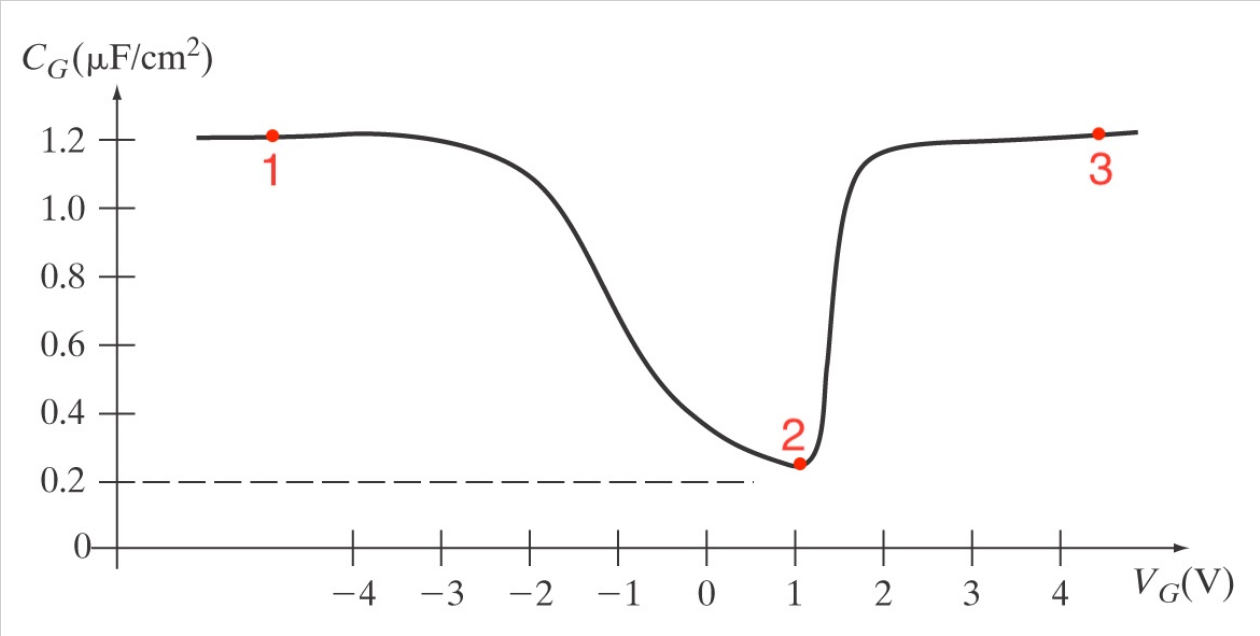
(correct: 5p, wrong: 0p, blank: 0p, minimum score 0p)

Majority carrier mobility: $\text{cm}^2/(\text{V}\cdot\text{s})$

(correct: 5p, wrong: 0p, blank: 0p, minimum score 0p)

Maximum marks: 10

10



In the following we will consider the capacitance-voltage relation for an n-channel MOS capactitor.

Fill in the blanks in the following two sections of text. Words are suggested for each blank – choose the most appropriate one. (correct: 1p, wrong: -0.5p, blank: -0.5p, minimum score: 0p)

The figure above shows the Select alternative (high, low) frequency C-V characteristics of an n-channel MOS capacitor.

When a strong negative bias voltage (point 1) is applied, the semiconductor surface has a

Select alternative (negative, positive)charge due to Select alternative (depletion of electrons, depletion of holes, accumulation of holes, accumulation of electrons) at the oxide-

semiconductor interface. In this region the semiconductor surface charge Select alternative (changes a lot with, is independent of, changes slightly with)surface potential, and the MOS

capacitance is equal to Select alternative (the depletion layer capacitance, the depletion layer and insulator capacitances in parallel, the insulator capacitance).

As the bias voltage increases/becomes less negative the capacitance decreases. For a small

positive bias voltage, the bands at the semiconductor surface bends Select alternative

(down, up), and the semiconductor surface becomes Select alternative (neutral, negative,

positive) due to Select alternative (accumulation of holes, accumulation of electrons, depletion of holes, depletion of electrons). At point 2 the capacitance is at its minimum and is

equal to Select alternative (the depletion layer capacitance, the depletion layer and insulator capacitances in series, the insulator capacitance).

When a large positive bias voltage is applied (point 3), the MOS capacitor is biased in

Select alternative (accumulation, depletion, inversion). At point 3 in the figure, the

capacitance is equal to Select alternative (the depletion layer and insulator capacitances in parallel, the insulator capacitance, the depletion layer capacitance), because

Select alternative (there is time for the charge in the inversion layer to change due to drift/diffusion of majority carriers, there is time for the charge in the inversion layer to change due to drift/diffusion of minority carriers, there is no time for the charge in the inversion layer to change in response to the gate voltage).

Maximum marks: 12

11 The threshold voltage of a MOSFET is given by

$$V_T = \Phi_{ms} - \frac{Q_i}{C_i} - \frac{Q_d}{C_i} + 2\phi_F \quad (1)$$

Which of the terms in (1) will depend on the doping concentration in the substrate?
(correct: 2p, wrong: -2p, blank: 0p, minimum score 0p)
Select one or more alternatives:

- ☐ $2\phi_F$
- ☐ $-\frac{Q_d}{C_i}$
- ☐ Φ_{ms}
- ☐ $-\frac{Q_i}{C_i}$

Maximum marks: 6

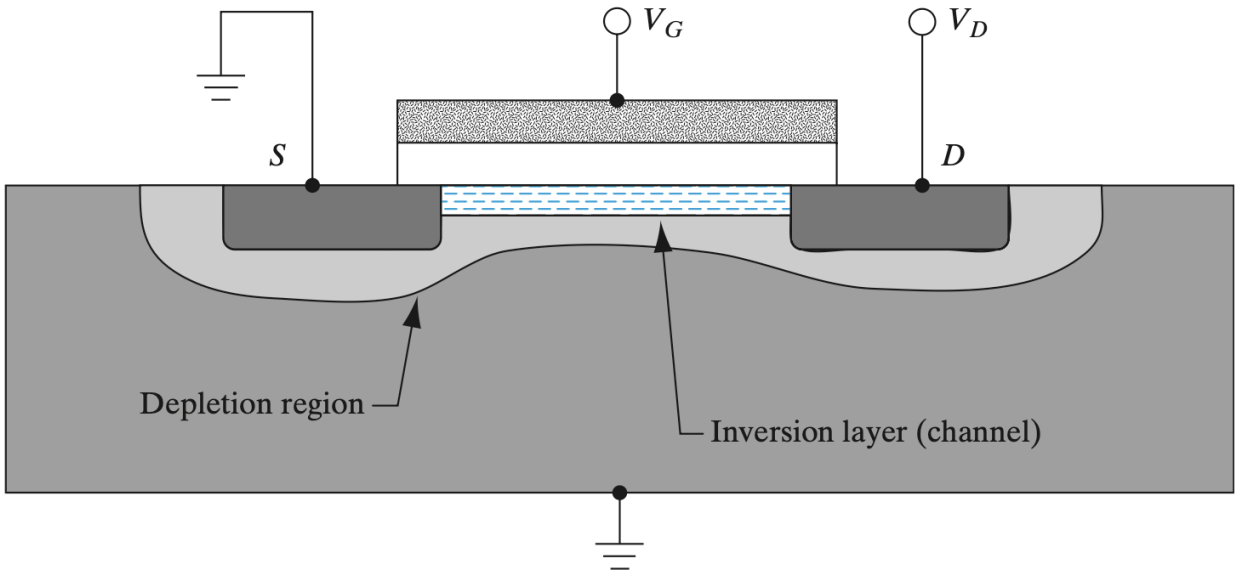
12 Consider a MOSFET where heavily doped n^+poly-silicon is used for the gate electrode, such that the energy level in the in the conduction band of the polysilicon gate, E_{cm}, equals the energy level in the conduction band of the semiconductor, E_{cs}.

What is true regarding the work function difference?
(correct: 2p, wrong: 0p, blank: 0p, minimum score 0p)
Select one alternative:

- ○ $\Phi_{ms} = 0$ for both n-channel and p-channel
- ○ $\Phi_{ms} < 0$ for both n-channel and p-channel
- ○ $\Phi_{ms} < 0$ for n-channel and $\Phi_{ms} > 0$ for p-channel
- ○ $\Phi_{ms} > 0$ for n-channel and $\Phi_{ms} < 0$ for p-channel

Maximum marks: 2

13



The figure above shows the cross section of an enhancement mode (normally off) n-channel MOSFET. Both the substrate and the source terminal is connected to ground with a potential of 0V.

Based on the indicated depletion region and inversion layer, select the correct voltage relationships.

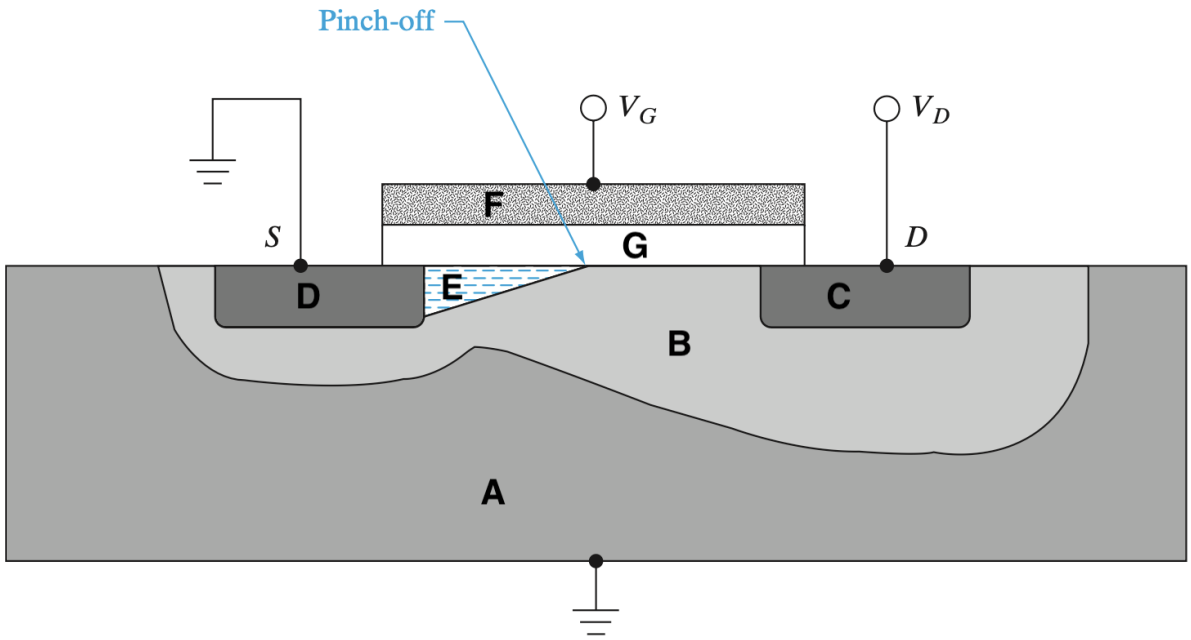
(correct: 2p, wrong: -2p, blank: 0p, minimum score 0p)

Select one or more alternatives:

- ☐ $V_D = V_G - V_T$
- ☐ $V_G > V_T$
- ☐ $V_D \approx 0\text{V}$
- ☐ $V_G < V_T$
- ☐ $V_G \approx 0\text{V}$

Maximum marks: 4

14



The figure above shows the cross section of an n-channel MOSFET. The different regions of the transistor are labeled with the letters A-F.

Which of the regions have a neutral space charge?

(correct: 0.5p, wrong: -0.5p, blank: 0p, minimum score 0p)

Select one or more alternatives:

A	F	G	E	C	D	B
☐	☐	☐	☐	☐	☐	☐

Which of the regions have substantial amount of free charge carriers?

(correct: 0.5p, wrong: -0.5p, blank: 0p, minimum score 0p)

Select one or more alternatives

G	D	C	A	B	F	E
☐	☐	☐	☐	☐	☐	☐

Which of the regions have strong electric fields across them?

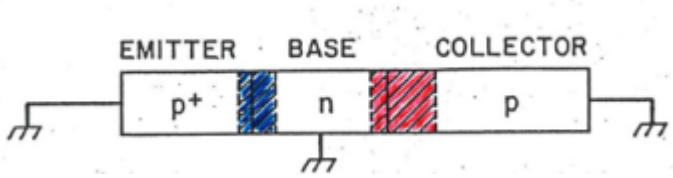
(correct: 0.5p, wrong: -0.5p, blank: 0p, minimum score 0p)

Select one or more alternatives

C	D	G	F	E	B	A
☐	☐	☐	☐	☐	☐	☐

Maximum marks: 6.5

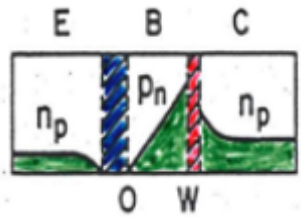
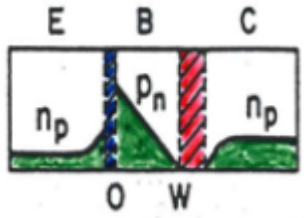
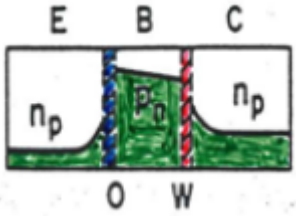
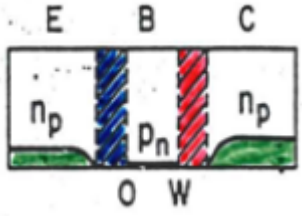
15



The figure above shows a PNP bipolar junction transistor in thermal equilibrium.
The figures below show the minority carrier distributions in the emitter (E), base (B), and collector (C), when the PNP transistor is operating in different regimes.

Match the figures and the names of the different operation regimes.
(correct: 1p, wrong: 0p, blank: 0p, minimum score 0p)

Please match the values:

	Cut-off	Normal active	Saturation	Inverted
	○	○	○	○
	○	○	○	○
	○	○	○	○
	○	○	○	○

Maximum marks: 4

16

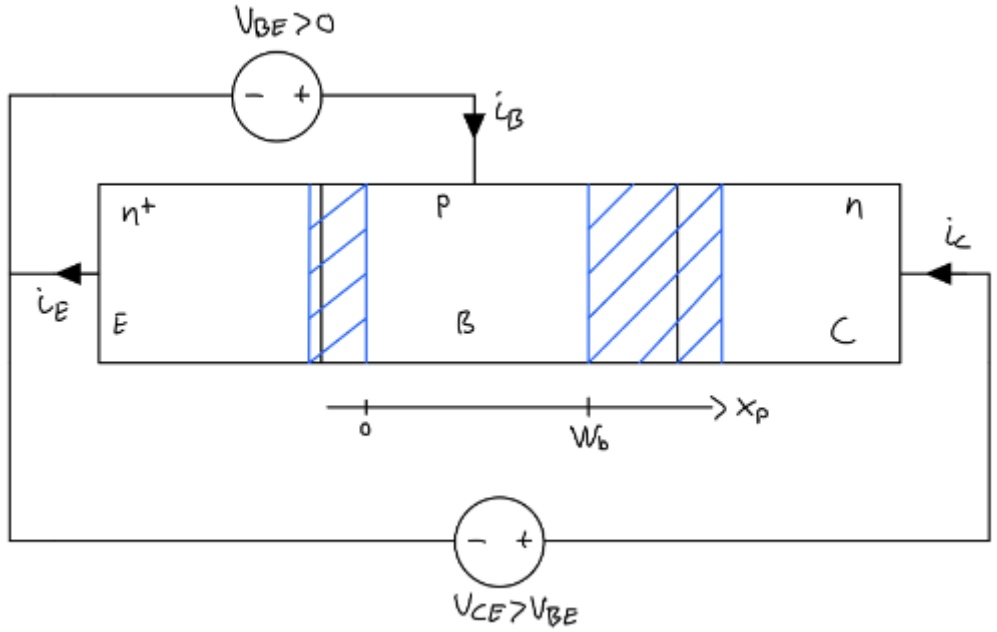


Figure 1: An NPN-transistor biased in normal active mode. The depletion regions are indicated in blue.

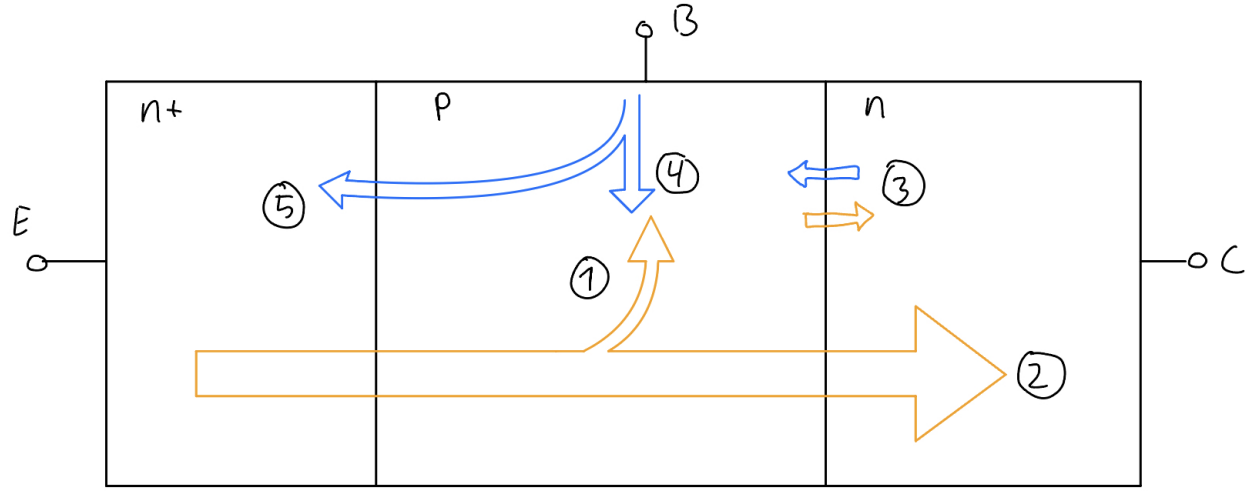


Figure 2: A sketch of the flow of holes and electrons within the NPN-transistor.

The text below explains the different current components in the NPN transistor shown in figure 2.

Fill in the blanks in the following two sections of text. Words are suggested for each blank – choose the most appropriate one. (correct: 1p, wrong: -0.5p, blank: -0.5p, minimum score: 0p)

In an NPN transistor operating in the normal active mode, electrons are injected from the emitter

to the base through the Select alternative (forward, reversed)biased $n^+ - p$ junction. This

Select alternative (minority, majority) carrier injection is giving rise to a distribution of

excess electrons in the Select alternative (positively charged, neutral, negatively charged) base region. Some of these electrons are lost in the base by recombination with holes (1). The

rest of the electrons will Select alternative (diffuse, drift) across the base region and reach

the Select alternative (forward, reverse) biased base-collector junction. The electrons

Select alternative (drift across the junction due to the strong electric field, diffuse across the junction because of the low potential barrier, recombine in the depletion region, emit photons) and leave the collector as the main part of the collector current (2).

In addition to the electrons that are injected from the emitter, Select alternative (doping

atoms, photons, electrons and holes) that are Select alternative (emitted, thermally generated, ionized) in the proximity of the base-collector junction will give rise to a

Select alternative (reverse, forward, breakdown, stimulated) current (3), adding to the total collector current. This contribution is usually very small compared to (2).

The base current must resupply the holes that are lost in the base due to recombination

with electrons in order to maintain Select alternative (the positive space charge, charge neutrality, the negative space charge). In addition to this recombination current (4), there will be

some holes flowing in the Select alternative (reverse, forward, sideways) biased emitter-base junction (5).

Maximum marks: 11

17 Which of the following currents are dominated by diffusion?
(correct: 2p, wrong: -2p, blank: 0p, minimum score 0p)
Select one or more alternatives:

- [ ] The minority carrier current in the neutral region of a p-n junction
- [ ] The majority carrier current in the neutral region of a p-n junction
- [ ] Electron-hole-pairs, optically generated within the depletion region of a photodiode, being separated by the junction field
- [ ] The drain current in a MOSFET
- [ ] The hole current in the base region of a p-n-p BJT

Maximum marks: 4

18

InsperaScan (20 p.)

At equilibrium, there can be no net hole or electron current in a p-n junction. Use the criteria
$J_n(x) = q\left[\mu_n n(x)\mathcal{E}(x) + D_n \frac{dn(x)}{dx}\right] = 0$
to show that the contact potential V_0 is related to the doping concentrations by
$V_0 = \frac{kT}{q} \ln \frac{N_a N_d}{n_i^2}$.

Maximum marks: 20

19

InsperaScan (20 p.)

Consider a reversed biased Si p-n junction with doping concentrations $N_A = N_D = 4 \cdot 10^{18}\mathrm{cm}^{-3}$. Let V_r denote the reverse bias voltage, $V_j = V_0 + V_r$ the total voltage difference across the p-n junction and let W denote the width of the depletion region.

Assume the critical field strength for Zener breakdown is $\mathcal{E}_B = 10^6 \mathrm{V/cm}$. Find the reverse bias voltage V_r for which Zener breakdown occurs, i.e. when the maximum value of the electric field reaches $\mathcal{E}_B$.

Given:
Relative permittivity in Si: $\epsilon_r = 11.8$
Permittivity in vacuum: $\epsilon_0 = 8.85 \cdot 10^{-14}\mathrm{F/cm}$
Intrinsic charge carrier density in Si: $n_i = 1.5 \cdot 10^{10}\mathrm{cm}^{-3}$.

Maximum marks: 20

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Attached



Formulas TFE-4146

SEMICONDUCTOR PHYSICS

Electron Momentum: $\mathrm{p} = m^*\mathrm{v} = \hbar\mathbf{k} = \dfrac{h}{\lambda}$ Planck: $E = h\nu = \hbar\omega$

Kinetic: $E = \dfrac{1}{2}m^*\mathrm{v}^2 = \dfrac{1}{2}\dfrac{\mathrm{p}^2}{m^*} = \dfrac{\hbar^2}{2m^*}\mathbf{k}^2$ (3-4) Effective mass: $m^* = \dfrac{\hbar^2}{d^2E/d\mathbf{k}^2}$ (3-3)

Total electron energy = $P.E. + K.E. = E_c + E(\mathbf{k})$

Fermi-Dirac e^- distribution: $f(E) = \dfrac{1}{e^{(E-E_F)/kT}+1} \cong e^{(E_F-E)/kT}$ for $E \gg E_F$ (3-10)

Equilibrium: $n_0 = \displaystyle\int_{E_c}^{\infty} f(E)N(E)dE = N_c f(E_c) = N_c e^{-(E_c-E_F)/kT}$ (3-15)

$N_c = 2\left(\dfrac{2\pi m_n^* kT}{h^2}\right)^{3/2}$ $N_v = 2\left(\dfrac{2\pi m_p^* kT}{h^2}\right)^{3/2}$ (3-16), (3-20)

$p_0 = N_v[1 - f(E_v)] = N_v e^{-(E_F-E_v)/kT}$ (3-19)

$n_i = N_c e^{-(E_c-E_i)/kT}$, $p_i = N_v e^{-(E_i-E_v)/kT}$ (3-21)

$n_i = \sqrt{N_c N_v}\, e^{-E_g/2kT} = 2\left(\dfrac{2\pi kT}{h^2}\right)^{3/2}(m_n^* m_p^*)^{3/4}e^{-E_g/2kT}$ (3-23), (3-26)

Equilibrium: $n_0 = n_i e^{(E_F-E_i)/kT}$, $p_0 = n_i e^{(E_i-E_F)/kT}$ (3-25) $n_0 p_0 = n_i^2$ (3-24)

Steady state: $n = N_c e^{-(E_c-F_n)/kT} = n_i e^{(F_n-E_i)/kT}$, $p = N_v e^{-(F_p-E_v)/kT} = n_i e^{(E_i-F_p)/kT}$ (4-15) $np = n_i^2 e^{(F_n-F_p)/kT}$ (5-38)

$\mathscr{E}(x) = -\dfrac{d\mathscr{V}(x)}{dx} = \dfrac{1}{q}\dfrac{dE_i}{dx}$ (4-26)

Poisson: $\dfrac{d\mathscr{E}(x)}{dx} = -\dfrac{d^2\mathscr{V}(x)}{dx^2} = \dfrac{\rho(x)}{\epsilon} = \dfrac{q}{\epsilon}(p - n + N_d^+ - N_a^-)$ (5-14)

$\mu \equiv \dfrac{q\bar{t}}{m^*}$ (3-40a) Drift: $\mathrm{v}_d \cong \dfrac{\mu\mathscr{E}}{1 + \mu\mathscr{E}/\mathrm{v}_s}\begin{cases} = \mu\mathscr{E} \text{ (low fields, ohmic)} \\ = \mathrm{v}_s \text{ (high fields, saturated vel.)} \end{cases}$ (Fig. 6-9)

Drift current density: $\dfrac{I_x}{A} = J_x = q(n\mu_n + p\mu_p)\mathscr{E}_x = \sigma\mathscr{E}_x$ (3-43)

$$J_n(x) = q\mu_n n(x)\mathscr{E}(x) + qD_n \frac{dn(x)}{dx}$$

Conduction Current: drift diffusion (4–23)

$$J_p(x) = q\mu_p p(x)\mathscr{E}(x) - qD_p \frac{dp(x)}{dx}$$

$$J_{\text{total}} = J_{\text{conduction}} + J_{\text{displacement}} = J_n + J_p + C\frac{dV}{dt}$$

Continuity: $\frac{\partial p(x,t)}{\partial t} = \frac{\partial \delta p}{\partial t} = -\frac{1}{q}\frac{\partial J_p}{\partial x} - \frac{\delta p}{\tau_p}$ $\frac{\partial \delta n}{\partial t} = \frac{1}{q}\frac{\partial J_n}{\partial x} - \frac{\delta n}{\tau_n}$ (4–31)

For steady state diffusion: $\frac{d^2\delta n}{dx^2} = \frac{\delta n}{D_n\tau_n} \equiv \frac{\delta n}{L_n^2}$ $\frac{d^2\delta p}{dx^2} = \frac{\delta p}{L_p^2}$ (4–34)

Diffusion length: $L \equiv \sqrt{D\tau}$ Einstein relation: $\frac{D}{\mu} = \frac{kT}{q}$ (4–29)

p-n JUNCTIONS

Equilibrium: $V_0 = \frac{kT}{q}\ln\frac{p_p}{p_n} = \frac{\text{kT}}{\text{q}}\ln\frac{N_a}{n_i^2/N_d} = \frac{\text{kT}}{\text{q}}\ln\frac{N_aN_d}{n_i^2}$ (5–8)

$\frac{p_p}{p_n} = \frac{n_n}{n_p} = e^{qV_0/kT}$ (5–10) $W = \left[\frac{2\epsilon(V_0 - V)}{q}\left(\frac{N_a + N_d}{N_aN_d}\right)\right]^{1/2}$ (5–57)

One-sided abrupt p^+-n: $x_{n0} = \frac{WN_a}{N_a + N_d} \simeq W$ (5–23) $V_0 = \frac{qN_dW^2}{2\epsilon}$

$\Delta p_n = p(x_{n0}) - p_n = p_n(e^{qV/kT} - 1)$ (5–29)

$\delta p(x_n) = \Delta p_n e^{-x_n/L_p} = p_n(e^{qV/kT} - 1)e^{-x_n/L_p}$ (5–31b)

Ideal diode: $I = qA\left(\frac{D_p}{L_p}p_n + \frac{D_n}{L_n}n_p\right)(e^{qV/kT} - 1) = I_0(e^{qV/kT} - 1)$ (5–36)

Non-ideal: $I = I_0'(e^{qV/\mathfrak{n}kT} - 1)$ ($\mathfrak{n}$ = 1 to 2) (5–74)

With light: $I_{\text{op}} = qAg_{\text{op}}(L_p + L_n + W)$ (8–1)

$$I_D = \frac{\bar{\mu}_n Z C_i}{L}\left\{(V_G - V_{FB} - 2\phi_F - \tfrac{1}{2}V_D)V_D - \frac{2}{3}\frac{\sqrt{2\epsilon_s q N_a}}{C_i}[(V_D + 2\phi_F)^{3/2} - (2\phi_F)^{3/2}]\right\} \quad (6\text{-}50)$$

$$I_D \simeq \frac{\bar{\mu}_n Z C_i}{L}[(V_G - V_T)V_D - \tfrac{1}{2}V_D^2] \quad (6\text{-}49)$$

Saturation: $I_D(\text{sat.}) \simeq \frac{1}{2}\bar{\mu}_n C_i \frac{Z}{L}(V_G - V_T)^2 = \frac{Z}{2L}\bar{\mu}_n C_i V_D^2(\text{sat.}) \quad (6\text{-}53)$

$$g_m = \frac{\partial I_D}{\partial V_G} \quad ; \quad g_m(\text{sat.}) = \frac{\partial I_D(\text{sat.})}{\partial V_G} \simeq \frac{Z}{L}\bar{\mu}_n C_i(V_G - V_T) \quad (6\text{-}54)$$

For short L: $I_D \simeq ZC_i(V_G - V_T)v_s \quad (6\text{-}60)$

Subthreshold slope: $S = \frac{dV_G}{d(\log I_D)} = \frac{kT}{q}\ln 10\left[1 + \frac{C_d + C_{it}}{C_i}\right] \quad (6\text{-}66)$

BJT-p-n-p

$$I_{Ep} = qA\frac{D_p}{L_p}\left(\Delta p_E \operatorname{ctnh}\frac{W_b}{L_p} - \Delta p_C \operatorname{csch}\frac{W_b}{L_p}\right) \quad (7\text{-}18) \qquad \begin{aligned}\Delta p_E &= p_n(e^{qV_{EB}/kT} - 1)\\ \Delta p_C &= p_n(e^{qV_{CB}/kT} - 1)\end{aligned} \quad (7\text{-}8)$$

$$I_C = qA\frac{D_p}{L_p}\left(\Delta p_E \operatorname{csch}\frac{W_b}{L_p} - \Delta p_C \operatorname{ctnh}\frac{W_b}{L_p}\right)$$

$$I_B = qA\frac{D_p}{L_p}\left[(\Delta p_E + \Delta p_C)\tanh\frac{W_b}{2L_p}\right] \quad (7\text{-}19)$$

$$B = \frac{I_C}{I_{Ep}} = \frac{\operatorname{csch} W_b/L_p}{\operatorname{ctnh} W_b/L_p} = \operatorname{sech}\frac{W_b}{L_p} \simeq 1 - \left(\frac{W_b^2}{2L_p^2}\right) \quad (7\text{-}26)$$

(Base transport factor)

$$\gamma = \frac{I_{Ep}}{I_{En} + I_{Ep}} = \left[1 + \frac{L_p^n n_n \mu_n^p}{L_n^p p_p \mu_p^n}\tanh\frac{W_b}{L_p^n}\right]^{-1} \simeq \left[1 + \frac{W_b n_n \mu_n^p}{L_n^p p_p \mu_p^n}\right]^{-1} \quad (7\text{-}25)$$

(Emitter injection efficiency)

$$\frac{i_C}{i_E} = B\gamma \equiv \alpha \quad (7\text{-}3) \qquad \frac{i_C}{i_B} = \frac{B\gamma}{1 - B\gamma} = \frac{\alpha}{1 - \alpha} \equiv \beta \quad (7\text{-}6) \qquad \frac{i_C}{i_B} = \beta = \frac{\tau_p}{\tau_t} \quad (7\text{-}7)$$

(Common base gain) (Common emitter gain) (For $\gamma = 1$)

Capacitance: $C = \left|\frac{dQ}{dV}\right|$ (5-55)

Junction Depletion: $C_j = \epsilon A\left[\frac{q}{2\epsilon(V_0 - V)}\frac{N_d N_a}{N_d + N_a}\right]^{1/2} = \frac{\epsilon A}{W}$ (5-62)

Stored charge
exp. hole dist.: $Q_p = qA\int_0^\infty \delta p(x_n)dx_n = qA\Delta p_n\int_0^\infty e^{-x_n/L_p}dx_n = qAL_p\Delta p_n$ (5-39)

$I_p(x_n = 0) = \frac{Q_p}{\tau_p} = qA\frac{L_p}{\tau_p}\Delta p_n = qA\frac{D_p}{L_p}p_n(e^{qV/kT} - 1)$ (5-40)

$G_s = \frac{dI}{dV} = \frac{qAL_p p_n}{\tau_p}\frac{d}{dV}(e^{qV/kT}) = \frac{q}{kT}I$ (5-67c)

Long p^+-n: $i(t) = \frac{Q_p(t)}{\tau_p} + \frac{dQ_p(t)}{dt}$ (5-47)

MOS-n CHANNEL

Oxide: $C_i = \frac{\epsilon_i}{d}$ Depletion: $C_d = \frac{\epsilon_s}{W}$ MOS: $C = \frac{C_i C_d}{C_i + C_d}$ (6-36)

Threshold: $V_T = \underbrace{\Phi_{ms} - \frac{Q_i}{C_i}}_{\text{Flat band}} - \frac{Q_d}{C_i} + 2\phi_F$ (6-38)

Inversion: $\phi_s\,(\text{inv.}) = 2\phi_F = 2\frac{kT}{q}\ln\frac{N_a}{n_i}$ (6-15) $W = \left[\frac{2\epsilon_s\phi_s}{qN_a}\right]^{1/2}$ (6-30)

$Q_d = -qN_aW_m = -2(\epsilon_s qN_a\phi_F)^{1/2}$ (6-32) At V_{FB}: $C_{FB} = \frac{C_i C_{\text{debye}}}{C_i + C_{\text{debye}}}$

Debye screening length: $L_D = \sqrt{\frac{\epsilon_s kT}{q^2 p_0}}$ (6-25) $C_{\text{debye}} = \frac{\epsilon_s}{L_D}$ (6-40)

Substrate bias: $\Delta V_T \simeq \frac{\sqrt{2\epsilon_s qN_a}}{C_i}(-V_B)^{1/2}$ (n channel) (6-63)

Document 3
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Physical Constants and Conversion Factors for the Exam in TFE-4146

Avogadro's number	$N_A = 6.02 \times 10^{23}$ molecules/mole
Boltzmann's constant	$k = 1.38 \times 10^{-23}$ J/K
	$= 8.62 \times 10^{-5}$ eV/K
Electronic charge (magnitude)	$q = 1.60 \times 10^{-19}$ C
Electronic rest mass	$m_0 = 9.11 \times 10^{-31}$ kg
Permittivity of free space	$\epsilon_0 = 8.85 \times 10^{-14}$ F/cm
	$= 8.85 \times 10^{-12}$ F/m
Planck's constant	$h = 6.63 \times 10^{-34}$ J-s
	$= 4.14 \times 10^{-15}$ eV-s
Room temperature value of kT	$kT = 0.0259$ eV
Speed of light	$c = 2.998 \times 10^{10}$ cm/s

1 Å (angstrom) $= 10^{-8}$ cm	
1 μm (micron) $= 10^{-4}$ cm	
1 nm = 10 Å $= 10^{-7}$ cm	
2.54 cm = 1 in.	
1 eV $= 1.6 \times 10^{-19}$ J	

Prefixes:

milli-,	m-	$= 10^{-3}$
micro-,	μ-	$= 10^{-6}$
nano-,	n-	$= 10^{-9}$
pico-,	p-	$= 10^{-12}$
kilo-,	k-	$= 10^{3}$
mega-,	M-	$= 10^{6}$
giga-,	G-	$= 10^{9}$

A wavelength λ of 1 μm corresponds to a photon energy of 1.24 eV.